

PHYSICS MODULE 3 MAGNETISM SOLVED PYQ

2025

Group A: Very Short Answer Type Questions

1. For which type of materials magnetic susceptibility is negative?

- **Answer: Diamagnetic Materials**

- **Detailed Explanation:**

Magnetic susceptibility (χ_m) is the measure of how much a material becomes magnetized in an applied magnetic field ($\vec{M} = \chi_m \vec{H}$).

- **Diamagnetic materials** (like copper, water, gold) create an induced magnetic field opposite to the external field, resulting in a small negative susceptibility ($\chi_m < 0$).
- Paramagnetic materials have positive susceptibility.

2. What is the dimension of $\mu_0 \epsilon_0$?

- **Answer: $L^{-2}T^2$**

- **Detailed Explanation:**

This relationship comes from Maxwell's electromagnetic wave equation, where the speed of light (c) is:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Rearranging for the constants:

$$\mu_0 \epsilon_0 = \frac{1}{c^2}$$

Since velocity (c) has dimensions LT^{-1} , the square is L^2T^{-2} .

The inverse is:

$$\frac{1}{[L^2T^{-2}]} = [L^{-2}T^2]$$

Group C: Long Answer Type Questions

(Note: These appeared as sub-parts of Question 7 in the original paper)

3. Is the magnetic field an irrotational vector or a solenoidal vector?

- **Answer: Solenoidal Vector**

- **Detailed Explanation:**

A vector field is **solenoidal** if its divergence is zero. According to Maxwell's equation (Gauss's law for magnetism):

$$\nabla \cdot \vec{B} = 0$$

This equation signifies that there are no magnetic monopoles; magnetic field lines always form continuous closed loops. Because the divergence is zero, the magnetic field is strictly solenoidal.

4. Is the electric field an irrotational vector or a solenoidal vector?

- **Answer: It depends (Electrostatic vs. Time-varying)**

- **Detailed Explanation:**

- **Electrostatic Case:** If charges are stationary, Maxwell's equations state $\nabla \times \vec{E} = 0$. A field with zero curl is **irrotational** (conservative).

- **Time-varying Case (Electrodynamics):** If there is a changing magnetic field,

Faraday's Law states $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$. Since the curl is not zero, the field is **not irrotational**.

- *Exam Tip:* If asked simply without context, it is often safer to specify: "Electrostatic fields are irrotational, but induced electric fields are not."

Paper: 2024

Group A: Very Short Answer Type Questions

5. The Curl of a conservative vector field is _____.

- **Answer: Zero (or a Null Vector)**

- **Detailed Explanation:**

A conservative vector field can always be expressed as the gradient of a scalar potential ($\vec{F} = \nabla \phi$). A fundamental identity in vector calculus states that the curl of a gradient is always zero:

$$\nabla \times (\nabla \phi) = 0$$

In electromagnetism, this relates to the electrostatic field, where $\nabla \times \vec{E} = 0$ (in the absence of changing magnetic fields).

6. The displacement current arises due to _____.

- **Answer: Time-varying Electric Field (or Changing Electric Flux)**

- **Detailed Explanation:**

Maxwell introduced the concept of displacement current (■) to resolve an

inconsistency in Ampere's Law for time-varying fields (like inside a charging capacitor). It depends on the rate of change of the electric flux (Φ_E):

$$I_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

7. Dimension of polarizability in SI units is _____.

- **Answer:** $M^{-1}T^4A^2$ (or equivalently expressed in Farad-meters squared, $F \cdot m^2$)
- **Detailed Explanation:**

Polarizability (α) connects the induced dipole moment (p) to the local electric field (E) via $p = \alpha E$.

- Unit of Dipole Moment (p): $C \cdot m$ (Coulomb-meter)
- Unit of Electric Field (E): V/m (Volt per meter) or N/C .

- Therefore, unit of $\alpha = \frac{C \cdot m}{V/m} = \frac{C \cdot m^2}{V}$.
- Since Capacitance $C = Q/V$, the unit is $F \cdot m^2$.
- In base SI dimensions:

$$[C] = A \cdot T$$

$$[V] = \frac{Work}{Charge} = \frac{ML^2T^{-2}}{AT} = ML^2T^{-3}A^{-1}$$

$$\alpha = \frac{[AT] \cdot [L^2]}{[ML^2T^{-3}A^{-1}]} = [M^{-1}L^0T^4A^2] = [M^{-1}T^4A^2]$$

Group C: Long Answer Type Questions

8. Dielectric Materials and Applications

(a) Give one example of Solid, Liquid and Gaseous Dielectric material.

- **Solid:** Mica, Glass, Porcelain, Ceramics.
- **Liquid:** Transformer Oil, Distilled Water.
- **Gas:** Air, Nitrogen (N_2), Sulfur Hexafluoride (SF_6).

(b) State three applications of Dielectric Materials.

1. **Capacitors:** To increase energy storage capacity ($C = \epsilon_r C_0$).
2. **Electrical Insulation:** Coating for wires and cables (e.g., PVC) to prevent short circuits.
3. **Transformers:** Transformer oil is used for both insulation and cooling.

(c) What is Dielectric Constant? What is its value in vacuum and conductor?

- **Definition:** The dielectric constant (ϵ_r or K) is the ratio of the permittivity of a substance to the permittivity of free space. It indicates how much the material reduces an external electric field.

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} = \frac{C}{C_0}$$

- **Value in Vacuum:** 1
- **Value in Conductor:** ∞ (Infinite, because the internal electric field is zero).

(d) Problem: Dielectric Calculation

- **Given:**
 - Relative permittivity (ϵ_r) = 4.2
 - Area (A) = 0.09 m^2
 - Capacitance (C) = $10 \times 10^{-9} \text{ F}$
 - Potential Difference (V) = 5 V
 - $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$
- **To Find:** Electric Field Strength (E).
- **Solution:**

The formula for a parallel plate capacitor is:

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

First, determine the separation distance (d):

$$d = \frac{\epsilon_0 \epsilon_r A}{C}$$

$$d = \frac{(8.854 \times 10^{-12}) \times (4.2) \times (0.09)}{10 \times 10^{-9}}$$

$$d \approx \frac{3.346 \times 10^{-12}}{10^{-5}} \approx 3.346 \times 10^{-7} \text{ m}$$

Now, calculate the Electric Field ($E = V/d$):

$$E = \frac{5}{3.346 \times 10^{-7}}$$

$$E \approx 1.49 \times 10^7 \text{ V/m}$$

9. Magnetic Properties and Materials

(a) Compare diamagnetic, paramagnetic and ferromagnetic materials.

Property	Diamagnetic	Paramagnetic	Ferromagnetic
Susceptibility (χ)	Small, Negative ($\chi < 0$)	Small, Positive ($\chi > 0$)	Very Large, Positive ($\chi \gg 0$)
Permeability (μ)	Slightly less than 1 ($\mu < \mu_0$)	Slightly greater than 1 ($\mu > \mu_0$)	Much greater than 1 ($\mu \gg \mu_0$)
Behavior in Field	Weakly repelled by magnets	Weakly attracted by magnets	Strongly attracted by magnets
Examples	Copper, Water, Gold	Aluminum, Platinum	Iron, Nickel, Cobalt

(b) Differentiate between Soft and Hard magnetic materials.

- **Soft Magnetic Materials:**
 - Low coercivity and retentivity.
 - Easily magnetized and demagnetized.
 - Narrow hysteresis loop (low energy loss).
 - **Use:** Transformers, temporary magnets (electromagnets).
- **Hard Magnetic Materials:**
 - High coercivity and retentivity.
 - Difficult to demagnetize.
 - Broad hysteresis loop (high energy loss).
 - **Use:** Permanent magnets (e.g., Alnico).

(c) Problem: Permeability Calculation

- **Given:**

1. Absolute Permeability (μ) = $0.12 \text{ N/A}^2 \cdot \text{H/m}$
 2. Permeability of free space (μ_0) = $4\pi \times 10^{-7} \approx 1.256 \times 10^{-6} \text{ N/A}^2 \cdot \text{H/m}$
- **To Find:** Relative Permeability (μ_r) and Susceptibility (χ_m).
 - **Solution:**

1. **Relative Permeability:**

$$\mu_r = \frac{\mu}{\mu_0} = \frac{0.12}{4\pi \times 10^{-7}} \approx \frac{0.12}{1.256 \times 10^{-6}}$$

$$\mu_r \approx 95,541$$

(This indicates a highly ferromagnetic material).

2. **Magnetic Susceptibility:**

$$\mu_r = 1 + \chi_m$$

$$\chi_m = \mu_r - 1 = 95,541 - 1 = 95,540$$

(d) Define magnetic domain of ferromagnetic materials.

- **Answer:** A magnetic domain is a microscopic region within a ferromagnetic material where the magnetic moments of individual atoms are aligned in the same direction due to strong exchange interactions.
 - In an unmagnetized state, the domains are randomly oriented, canceling each other out.
 - When an external field is applied, domains aligned with the field grow (domain wall displacement) or rotate, leading to strong magnetization.

Paper: 2023

Group A: Very Short Answer Type Questions

10. A conductor of length L is placed in a magnetic field of magnetic flux density B . If current I is flowing through the conductor then how much force will be experienced by it?

- **Answer:** $\vec{F} = I(\vec{L} \times \vec{B})$
- **Detailed Explanation:**
The force is the magnetic Lorentz force on a current-carrying wire. The magnitude is $F = BIL \sin \theta$, where θ is the angle between the length vector (direction of current) and the magnetic field. The direction is given by Fleming's Left-Hand Rule.

Group B: Short Answer Type Questions

(Note: These questions involve Vector Calculus, which is the mathematical foundation for Maxwell's Equations in this module)

11. What do you mean by irrotational vector? Show that

$\vec{A} = (6xy + z^3)\hat{i} + (3x^2 - z)\hat{j} + (3xz^2 - y)\hat{k}$ is irrotational.

- **Definition:** A vector field is **irrotational** if its curl is zero ($\nabla \times \vec{A} = 0$). This implies the field is conservative (like an electrostatic field).
- **Solution:**

We calculate the curl of $\vec{A}$:

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 6xy + z^3 & 3x^2 - z & 3xz^2 - y \end{vmatrix}$$

○ **i component:** $\frac{\partial}{\partial y}(3xz^2 - y) - \frac{\partial}{\partial z}(3x^2 - z) = (-1) - (-1) = 0$

○ **j component:** $\frac{\partial}{\partial z}(6xy + z^3) - \frac{\partial}{\partial x}(3xz^2 - y) = (3z^2) - (3z^2) = 0$

○ **k component:** $\frac{\partial}{\partial x}(3x^2 - z) - \frac{\partial}{\partial y}(6xy + z^3) = (6x) - (6x) = 0$

Since $\nabla \times \vec{A} = 0$, the vector is irrotational.

12. Evaluate

$$\vec{\nabla} \cdot \left(\frac{\vec{r}}{r^3} \right)$$

- **Answer: Zero** (for $r \neq 0$)
- **Detailed Explanation:**

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r^3 = (x^2 + y^2 + z^2)^{3/2}$.

Using the divergence identity $\nabla \cdot (f\vec{A}) = f(\nabla \cdot \vec{A}) + \vec{A} \cdot (\nabla f)$:

Here $f = \frac{1}{r^3}$ and $\vec{A} = \vec{r}$.

1. $\nabla \cdot \vec{r} = 3$

2. $\nabla \left(\frac{1}{r^3} \right) = -3r^{-4} \nabla r = -3r^{-4} \left(\frac{\vec{r}}{r} \right) = -\frac{3\vec{r}}{r^5}$

3. Combine: $\frac{1}{r^3}(3) + \vec{r} \cdot \left(-\frac{3\vec{r}}{r^5} \right) = \frac{3}{r^3} - \frac{3(\vec{r} \cdot \vec{r})}{r^5} = \frac{3}{r^3} - \frac{3r^2}{r^5} = \frac{3}{r^3} - \frac{3}{r^3} = 0$.

13. Magnetism and Hysteresis

(a) Define Magnetic susceptibility and permeability. Obtain the relation between them.

- **Susceptibility (χ_m):** The ratio of magnetization (M) to magnetic intensity (H).

$$\chi_m = M/H.$$

- **Permeability (μ):** The measure of a material's ability to support a magnetic field.

$$\mu = B/H.$$

- **Relation:**

$$B = \mu_0(H + M)$$

$$B = \mu_0 H \left(1 + \frac{M}{H}\right) = \mu_0 H (1 + \chi_m)$$

Since $B = \mu H$, we get $\mu = \mu_0(1 + \chi_m)$ or $\mu_r = 1 + \chi_m$.

(b) Write the differences between diamagnetic, paramagnetic and ferromagnetic substances.

- (Refer to Question 9(a) in the previous response for the comparison table).

(c) What is hysteresis? Draw a hysteresis curve for a sample of iron. Explain the curve step by step.

- **Hysteresis:** The lagging of magnetization (M) behind the magnetizing field (H) when a ferromagnetic material is subjected to cycles of magnetization.
- **Curve Explanation:**
 1. **O to A (Saturation):** As H increases, M increases until all domains align (Saturation point A).
 2. **A to B (Retentivity):** When H is reduced to 0, M does not drop to 0. The remaining magnetism (M_r) is **Retentivity**.
 3. **B to C (Coercivity):** To remove the residual magnetism, a negative field is applied. The field required to make $B = 0$ (H_c) is **Coercivity**.

14. Dielectrics and Capacitor Problem

(Question 9 in the PDF)

(a) What are dielectrics? Differentiate between polar and non-polar dielectrics.

- **Dielectrics:** Insulating materials that can be polarized by an electric field.
- **Polar:** Molecules have a permanent dipole moment (e.g., H_2O , HCl). Center of positive and negative charges do not coincide.
- **Non-polar:** No permanent dipole moment (e.g., CH_4 , CO_2). Centers coincide until an external field is applied.

(b) Find the relation between electronic polarizability and atomic radius.

- For a neutral atom of radius R in an electric field E , the induced dipole moment is $p = \alpha_e E$.
- The relation is derived as: $\alpha_e = 4\pi\epsilon_0 R^3$.

(c) Establish the relation $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$.

- This is the fundamental definition of Electric Displacement $\vec{D}$, combining the free space field ($\epsilon_0 \vec{E}$) and the material polarization ($\vec{P}$).

(d) Problem: Capacitor Calculations

- Given: $V = 10V$, $A = 6.45 \times 10^{-4} m^2$, $d = 0.002m$, $\epsilon_r = 6$.
- To Calculate:

1. Charge (Q):

$$C = \frac{\epsilon_0 \epsilon_r A}{d} = \frac{(8.854 \times 10^{-12})(6)(6.45 \times 10^{-4})}{0.002} \approx 1.71 \times 10^{-11} F$$

$$Q = CV = (1.71 \times 10^{-11})(10) = 1.71 \times 10^{-10} C$$

2. Net Polarization (P):

$$E = \frac{V}{d} = \frac{10}{0.002} = 5000 V/m$$

$$P = \epsilon_0 (\epsilon_r - 1) E = (8.854 \times 10^{-12})(5)(5000)$$

$$P \approx 2.21 \times 10^{-7} C/m^2$$

15. Maxwell's Equations and EM Waves

(Question 11 in the PDF)

(a) Write Maxwell's wave equations... prove velocity is c .

- From Maxwell's equations in vacuum, taking the curl of the curl equations leads to:

$$\nabla^2 \vec{E} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

Comparing this to the standard wave equation $\nabla^2 \psi - \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} = 0$, we get

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = c$$

(b) Problem: Electric Wave

- Given: $E = E_0 \cos(1000x - 50000t)$. Compare to $E = E_0 \cos(kx - \omega t)$.

- $k = 1000 \text{ m}^{-1}, \omega = 5000 \text{ rad/s}$.
- Wavelength (λ): $\frac{2\pi}{k} = \frac{2\pi}{1000} \approx 0.00628 \text{ m}$.
- Speed (v): $\frac{\omega}{k} = \frac{5000}{1000} = 5 \text{ m/s}$.

(c) Displacement Current vs Conduction Current:

- For a conductor with conductivity σ and permittivity ϵ in a field $E = E_0 e^{i\omega t}$:

$$J_c = \sigma E$$

$$J_d = \epsilon \frac{\partial E}{\partial t} = i\omega \epsilon E$$

$$\frac{|J_d|}{|J_c|} = \frac{\omega \epsilon}{\sigma}$$

Ratio $\frac{|J_d|}{|J_c|} = \frac{\omega \epsilon}{\sigma}$. At low frequencies for good conductors, this ratio is very small, meaning J_d is negligible.

(d) Problem: Rotating Disc (EM Induction)

- **Given:** $r = 0.05 \text{ m}, \omega = 1000 \text{ rpm} = \frac{100\pi}{3} \text{ rad/s}$. EMF = $3.22 \times 10^{-3} \text{ V}$.

- **Formula:** $EMF = \frac{1}{2} B \omega r^2$.
- **Solution:**

$$B = \frac{2 \times EMF}{\omega r^2} = \frac{2 \times 0.00322}{\left(\frac{100\pi}{3}\right) \times (0.05)^2}$$

$$B \approx 0.0246 \text{ Tesla}$$

Paper: 2019

Group A: Multiple Choice Type Questions

16. The value of n for which $\vec{A} = 2ax\hat{i} + 2y\hat{j} + 4z\hat{k}$ is solenoidal is equal to:

- **Answer: (c) -3**
- **Detailed Explanation:**

A vector is solenoidal if its divergence is zero ($\nabla \cdot \vec{A} = 0$).

$$\nabla \cdot \vec{A} = \frac{\partial}{\partial x}(2ax) + \frac{\partial}{\partial y}(2y) + \frac{\partial}{\partial z}(4z)$$

$$= 2a + 2 + 4 = 2a + 6$$

For solenoidal condition: $2a + 6 = 0 \Rightarrow 2a = -6 \Rightarrow a = -3$.

17. In a conservative field:

- **Answer: (b) line integral is independent of path**
- **Detailed Explanation:**

A conservative field (like the Electrostatic field) can be expressed as the gradient of a potential. This implies that the work done moving between two points depends only on the endpoints, not the path taken.

- (a) Work done is zero *only for a closed path*.
- (c) Curl is zero (irrotational).
- (d) Divergence is not necessarily zero.

18. Clausius-Mossotti equation can be expressed as:

- **Answer: (a)** $\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha_e}{3\epsilon_0}$
- **Detailed Explanation:**

This equation relates the macroscopic dielectric constant (ϵ_r) to the microscopic polarizability (α_e) of the atoms/molecules in the material.

- Note: The option in the paper uses ϵ (likely a typo for ϵ_r) and equates it to

$$\frac{N\alpha_e}{3\epsilon_0}.$$

19. Material which shows negative susceptibility is:

- **Answer: (b) Diamagnetic**
- **Detailed Explanation:**

Diamagnetic materials create an induced magnetic field opposite to the external field, resulting in a negative magnetic susceptibility ($\chi < 0$).

Group B: Short Answer Type Questions

20. Dielectric Relations

(a) Establish the relation $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$ where the parameters have their usual meaning.

- **Answer:**

Consider a dielectric material in an external electric field $\vec{E}$. The total electric flux density ($\vec{D}$) accounts for both free charge and bound (polarized) charge.

1. Gauss's Law in free space: $\nabla \cdot \vec{E} = \rho_{total} / \epsilon_0$.
2. $\rho_{total} = \rho_{free} + \rho_{bound}$.
3. Bound charge density is related to polarization: $\rho_{bound} = -\nabla \cdot \vec{P}$.
4. Substituting back: $\epsilon_0 \nabla \cdot \vec{E} = \rho_{free} - \nabla \cdot \vec{P}$.
5. $\nabla \cdot (\epsilon_0 \vec{E} + \vec{P}) = \rho_{free}$.
6. Define $\vec{D} = \epsilon_0 \vec{E} + \vec{P}$, such that $\nabla \cdot \vec{D} = \rho_{free}$.

(b) Distinguish between nonpolar and polar molecules. Give examples.

- **Non-Polar Molecules:**
 - The center of positive charge coincides with the center of negative charge.
 - No permanent dipole moment.
 - **Examples:** $H_2, O_2, N_2, CO_2, CH_4$.
- **Polar Molecules:**
 - The centers of positive and negative charge are separated.
 - Have a permanent dipole moment.
 - **Examples:** H_2O, HCl, NH_3 .

Group C: Long Answer Type Questions

21. Maxwell's Equations and Magnetism

(Question 8 in the PDF)

(a) Write down Maxwell's wave equation for free space with physical significance and from there prove that velocity of EM wave is velocity of light.

- **Wave Equation:**

$$\nabla^2 \vec{E} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} = 0$$

$$\nabla^2 \vec{B} - \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} = 0$$

- **Physical Significance:** These equations predict that electric and magnetic fields can propagate through space as waves, decoupled from charges or currents.
- **Velocity Proof:**

Comparing the equation to the standard wave equation $\nabla^2 y = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}$, we identify:

$$v = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

Substituting $\mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$ and $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$:

$$v \approx 3 \times 10^8 \text{ m/s} = c$$

(b) Write down the Maxwell's modification of Ampere's law.

- **Original Ampere's Law:** $\nabla \times \vec{B} = \mu_0 \vec{J}$ (valid only for steady currents).
- **Maxwell's Modification:** Added the "Displacement Current" term to account for time-varying electric fields (e.g., inside a capacitor).
- **Modified Equation:**

$$\nabla \times \vec{B} = \mu_0 \left(\vec{J}_{\text{conduction}} + \vec{J}_{\text{displacement}} \right)$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$

(c) Derive molecular field theory on ferromagnetism.

- **Weiss Molecular Field Theory:**
Weiss proposed that in ferromagnetic materials, there exists a strong internal "molecular field" (B_m) proportional to the magnetization (M):
$$B_{\text{eff}} = B_{\text{ext}} + B_m = B_{\text{ext}} + \lambda M$$
where λ is the Weiss constant.
- **Curie-Weiss Law:**
Using this effective field in the paramagnetic susceptibility formula, we derive:

$$\chi_m = \frac{C}{T - T_C}$$

where T_C is the **Curie Temperature**. Above T_C , the material behaves as a paramagnet; below T_C , it is ferromagnetic.

Paper: 2012-2013

Group A: Multiple Choice Questions

22. Electromagnetic wave is propagated through a region of vacuum... If the electric vector is given by $\vec{E} = E_0 \exp[i(kx - \omega t)]$, then the magnetic vector is:

- **Answer: (c) in z direction**
- **Detailed Explanation:**

In an electromagnetic wave, the propagation vector ($\hat{k}$), electric field ($\vec{E}$), and

magnetic field ($\vec{B}$) form a right-handed orthogonal triad.

- Propagation is in $\hat{k}$ direction (from $\vec{E} \times \vec{B}$).
- Electric field is in $\hat{i}$ direction ($\vec{E}$).
- Direction of $\vec{B}$ is along $\hat{k} \times \vec{E} = \hat{j} \times \hat{i} = \hat{k}$. Thus, $\vec{B}$ is in the **z-direction**.

23. At a point just outside a current carrying conducting wire:

- **Answer: (a) electric field is zero while the magnetic field is non-zero**
- **Detailed Explanation:**
 - A current-carrying wire is electrically neutral (equal number of protons and electrons), so the external **Electric Field is zero**.
 - However, the moving charges create a **Magnetic Field** (Oersted's experiment/Ampere's Law).

Group B: Short Answer Type Questions

24. Ampere's Law and Steady Current

(Question 4 in the PYQ)

(a) State Ampere's law in magnetostatics in integral form and from that deduce its differential form.

- **Integral Form:** $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$
- **Derivation:**

Using Stokes' Theorem,

$$\oint \vec{B} \cdot d\vec{l} = \int (\nabla \times \vec{B}) \cdot d\vec{s}$$

Also, $I_{\text{enclosed}} = \int \vec{J} \cdot d\vec{s}$

Equating integrals:

$$\int (\nabla \times \vec{B}) \cdot d\vec{s} = \mu_0 \int \vec{J} \cdot d\vec{s}$$

Differential Form: $\nabla \times \vec{B} = \mu_0 \vec{J}$

(b) Write down the condition of steady state current. Show that Ampere's law implies that the current is in the steady state.

- **Condition:** Steady current means charge density is constant ($\frac{\partial \rho}{\partial t} = 0$), implying $\nabla \cdot \vec{J} = 0$ (Continuity Equation).

- **Proof:** Taking the divergence of Ampere's law ($\nabla \times \vec{B} = \mu_0 \vec{J}$):

$$\nabla \cdot (\nabla \times \vec{B}) = \mu_0 (\nabla \cdot \vec{J})$$

Since the divergence of a curl is always zero, $0 = \mu_0 (\nabla \cdot \vec{J})$, which implies $\nabla \cdot \vec{J} = 0$ (Steady State). *Note: This inconsistency for time-varying fields led Maxwell to add the displacement current.*

Group C: Long Answer Type Questions

25. Lorentz Force Problem

(Question 7d in the PDF)

Problem: A test charge $q = 0.4\text{C}$ moves with velocity $\vec{v} = 4\hat{i} - \hat{j} + 2\hat{k}$ m/s through an Electric field $\vec{E} = 10\hat{i} + 10\hat{k}$ and Magnetic field $\vec{B} = 2\hat{i} - \hat{j} - 6\hat{k}$. Find the Lorentz force.

- **Formula:** $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$
- **Step 1: Calculate Magnetic Force ($\vec{v} \times \vec{B}$):**

$$\vec{v} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -1 & 2 \\ 2 & -1 & -6 \end{vmatrix}$$

$$= \hat{i}(6 - (-2)) - \hat{j}(-24 - 4) + \hat{k}(-4 - (-2))$$

$$= 8\hat{i} + 28\hat{j} - 2\hat{k}$$

- **Step 2: Add Electric Field:**

$$\vec{E} + (\vec{v} \times \vec{B}) = (10\hat{i} + 10\hat{k}) + (8\hat{i} + 28\hat{j} - 2\hat{k})$$

$$= 18\hat{i} + 28\hat{j} + 8\hat{k}$$

- **Step 3: Multiply by charge (q):**

$$\vec{F} = 0.4(18\hat{i} + 28\hat{j} + 8\hat{k}) = 7.2\hat{i} + 11.2\hat{j} + 3.2\hat{k} \text{ Newtons}$$

26. Vector Potential Calculation

(Question 9b in the PDF)

Problem: If the vector potential is $\vec{A} = (x^2 + y^2 - z^2)\hat{z}$, find the magnetic field $\vec{B}$ at $(1, 1, 1)$.

- **Formula:** $\vec{B} = \nabla \times \vec{A}$
- **Calculation:**

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial_x & \partial_y & \partial_z \\ 0 & x^2 + y^2 - z^2 & 0 \end{vmatrix}$$

- $\hat{i}$ -component: $-\frac{\partial}{\partial z}(x^2 + y^2 - z^2) = -(-2z) = 2z$
 - $\hat{j}$ -component: 0
 - $\hat{k}$ -component: $\frac{\partial}{\partial x}(x^2 + y^2 - z^2) = 2x$
- $$\vec{B} = 2z\hat{i} + 2x\hat{k}$$

- At point (1,1,1): $\vec{B} = 2(1)\hat{i} + 2(1)\hat{k} = 2\hat{i} + 2\hat{k}$.

27. Conservative Field Check

(Question 9a in the PDF)

Problem: Find if work done in field $\vec{F} = (2xy + z^3)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$ is independent of path.

- **Method:** Check if $\nabla \times \vec{F} = 0$ (Conservative).
- **Calculation:**
 - $\hat{i}$: $\partial_y(3xz^2) - \partial_z(x^2) = 0 - 0 = 0$
 - $\hat{j}$: $\partial_z(2xy + z^3) - \partial_x(3xz^2) = 3z^2 - 3z^2 = 0$
 - $\hat{k}$: $\partial_x(x^2) - \partial_y(2xy + z^3) = 2x - 2x = 0$
- **Result:** Since Curl is zero, the field is conservative, and **Work Done is independent of the path.**

28. Solenoid Magnetic Field

(Question 9d in the PDF)

Problem: Find the magnetic induction B on the axis of an infinitely long solenoid carrying current I with n turns per unit length.

- **Answer:** $B = \mu_0 n I$.
- **Derivation Outline:** Apply Ampere's Law to a rectangular loop with one side inside the solenoid and one side outside.

Paper: 2011 (Old Syllabus)

Group A: Multiple Choice Questions

29. The significance of $\nabla \cdot \vec{B} = 0$ ($\vec{B}$ is the magnetic field of induction) is that:

- Answer: (b) magnetic monopole cannot exist
- Detailed Explanation:

The divergence of a vector field represents a "source" or "sink."

- Gauss's Law for electricity ($\nabla \cdot \vec{E} = \rho/\epsilon_0$) shows that electric charges (monopoles) exist.
- Gauss's Law for magnetism ($\nabla \cdot \vec{B} = 0$) shows there are no magnetic charges (monopoles). Magnetic field lines always form closed loops.

30. Displacement current through an ideal capacitor:

- Answer: (c) is equal to conduction current
- Detailed Explanation:

Inside a circuit, the conduction current (I_c) flowing through the wires is continuous with the displacement current (I_d) flowing across the dielectric gap of the capacitor.

- $I_c = I_d = \epsilon_0 \frac{d\Phi_E}{dt}$.
- This continuity ensures Ampere's Law holds for the entire circuit.

31. The force of attraction between two long parallel current carrying wires separated by distance r is:

- Answer: (a) $\frac{\mu_0 I_1 I_2}{2\pi r}$
 - (Note: The question asks for Force per unit length, but the options are labeled as "B". Option (a) represents the correct form for Force per unit length $\frac{\mu_0 I_1 I_2}{2\pi r}$. If interpreted strictly as B-field, it would be $\frac{\mu_0 I}{2\pi r}$. Given the context, (a) is the intended relationship for interaction).

32. If $\phi = 1/r$, the value of $\nabla^2 \phi$ will be:

- Answer: (b) $-\vec{r}/r^3$
- Detailed Explanation:

$$\phi = (x^2 + y^2 + z^2)^{-1/2}.$$

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{i} + \dots = -\frac{1}{2}(x^2 + y^2 + z^2)^{-3/2}(2x)\hat{i} + \dots$$

$$= -\frac{x}{r^3}\hat{i} - \frac{y}{r^3}\hat{j} - \frac{z}{r^3}\hat{k} = -\frac{\vec{r}}{r^3}.$$

33. Which of the following is not a scalar field?

- **Answer: (a) Displacement of mosquito in space**
- **Detailed Explanation:**
 - Displacement is a **Vector** quantity (magnitude + direction).
 - Light intensity, Temperature, and Humidity are all **Scalar** quantities (magnitude only).

Group B: Short Answer Type Questions

34. Gauss's Law in Electrostatics

(Question 4 in the PDF)

- **State and prove Gauss's law:** The total electric flux through a closed surface is equal to $1/\epsilon_0$ times the enclosed charge ($\oint \vec{E} \cdot d\vec{s} = q_{enc}/\epsilon_0$).
- **Differential Form:** Apply Divergence Theorem ($\oint \vec{E} \cdot d\vec{s} = \int (\nabla \cdot \vec{E}) dV$) and equate to $\int (\rho/\epsilon_0) dV$.
Result: $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$.

35. Magnetic Field Calculation

(Question 5a in the PDF)

- **Problem:** Show that for a square loop of edge a carrying current I , the magnetic field at the center is $B = \frac{2\sqrt{2}\mu_0 I}{\pi a}$.
- **Hint:** Use the Biot-Savart law for a finite straight wire (length a , distance $a/2$) four times.

36. Vector Potential Problem

(Question 5b in the PDF)

- **Problem:** If vector potential $A = (x^2 + y^2 - z^2)\hat{j}$, find B at $(1, 1, 1)$.
(Note: This is identical to Question 26 from the 2012 paper).
- **Answer:** $B = 2\hat{i} + 2\hat{k}$.

Group C: Long Answer Type Questions

37. Solenoidal Vector

(Question 12c in the PDF)

- **Question:** When is a vector called solenoidal?
- **Answer:** A vector A is solenoidal if its divergence is zero ($\nabla \cdot A = 0$). This implies the field lines form closed loops and have no source or sink (e.g., Magnetic Field).

38. Ampere's Law and Continuity

(Question 11 in the PYQ)

- **Define Current Density:** Current per unit area perpendicular to flow ($\vec{J} = I/A_n$).
- **Equation of Continuity:** $\nabla \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0$. (Conservation of Charge).
- **Ampere's Law Differential Form:** $\nabla \times \vec{B} = \mu_0 \vec{J}$.

39. Magnetic Force on Electron

(Question 11 in the PYQ)

- **Problem:** Electron energy = 1000 eV, Magnetic field $B = 1000$ T. Find radius of circle.

- **Formula:** $r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$.
- **Calculation:**

$$K = 1000 \text{ eV} = 1000 \times 1.6 \times 10^{-19} \text{ J}.$$

$$m = 9.1 \times 10^{-31} \text{ kg}.$$

$$q = 1.6 \times 10^{-19} \text{ C}.$$

Calculate r using the values.

40. Electric Dipole

(Question 10a in the PDF)

- **Derivation:** Derive expression for Electric Field ($\blacksquare$) and Potential ($\blacksquare$) for an electric dipole at an external point (r, θ) .

$$* \quad V(r, \theta) = \frac{p \cos \theta}{4\pi\epsilon_0 r^2}.$$

$$\circ \quad E_r = -\frac{\partial V}{\partial r}, \quad E_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta}.$$

Paper: 2010-2011 (Old Syllabus)

Group A: Multiple Choice Questions

41. $\nabla \cdot \vec{B} = 0$ signifies that:

- **Answer: (b) magnetic monopole does not exist**
- **Explanation:** The divergence of a magnetic field is always zero, meaning magnetic field lines are continuous loops without a starting or ending point (source/sink).

42. Ampere's circuital law is valid for:

- **Answer: (b) steady current only**
- **Explanation:** The original form $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$ fails for time-varying fields (where displacement current is needed).

43. Electric field is:

- **Answer: (d) irrotational if the magnetic field does not change with time**
- **Explanation:** $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$. If $\vec{B}$ is constant (electrostatics), curl is zero (irrotational). If $\vec{B}$ changes, curl is non-zero.

44. A point charge $\blacksquare$ is placed at the centre of a cube of side $\blacksquare$. Flux through each surface?

- **Answer: (b) $q/6\epsilon_0$**
- **Explanation:** Total flux is q/ϵ_0 (Gauss's Law). A cube has 6 symmetric faces, so each face gets $q/6$ th.

Group B: Short Answer Type Questions

45. Vector Identities & Potentials

- **Question:** If $\vec{B} = \nabla \times \vec{A}$, show that $\nabla \cdot \vec{B} = 0$.
- **Proof:** Divergence of a Curl is always zero: $\nabla \cdot (\nabla \times \vec{A}) = 0$.
- **Question:** If $\vec{E}$ is electrostatic, show it is derived from a scalar potential.
- **Proof:** Since $\nabla \times \vec{E} = 0$, vector calculus states $\vec{E}$ can be written as the gradient of a scalar ϕ : $\vec{E} = -\nabla \phi$.

46. Ampere's Law Forms

- **Question:** State Ampere's law in integral and differential forms. Show it implies steady current.
- **Answer:**
 - Integral: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I$.
 - Differential: $\nabla \times \vec{B} = \mu_0 \vec{J}$.
 - Steady Current Proof: Take divergence of both sides. $\nabla \cdot (\nabla \times \vec{B}) = 0$, so $\mu_0 (\nabla \cdot \vec{J}) = 0 \Rightarrow \nabla \cdot \vec{J} = 0$.

Group C: Long Answer Type Questions

47. Maxwell's Equations & Wave Equation

- **Question:** State Maxwell's equations. Show that in a charge-free, current-free region, $\vec{E}$ satisfies a wave equation. Identify the velocity.
- **Answer:**
 - Start with $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ and $\nabla \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$ (vacuum).
 - Take curl of the first eq: $\nabla \times (\nabla \times \vec{E}) = -\frac{\partial}{\partial t} (\nabla \times \vec{B})$.
 - Use identity $\nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E}$. Since charge-free, $\nabla \cdot \vec{E} = 0$.
 - Result: $\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$.
 - Velocity $v = 1/\sqrt{\mu_0 \epsilon_0} = c$.

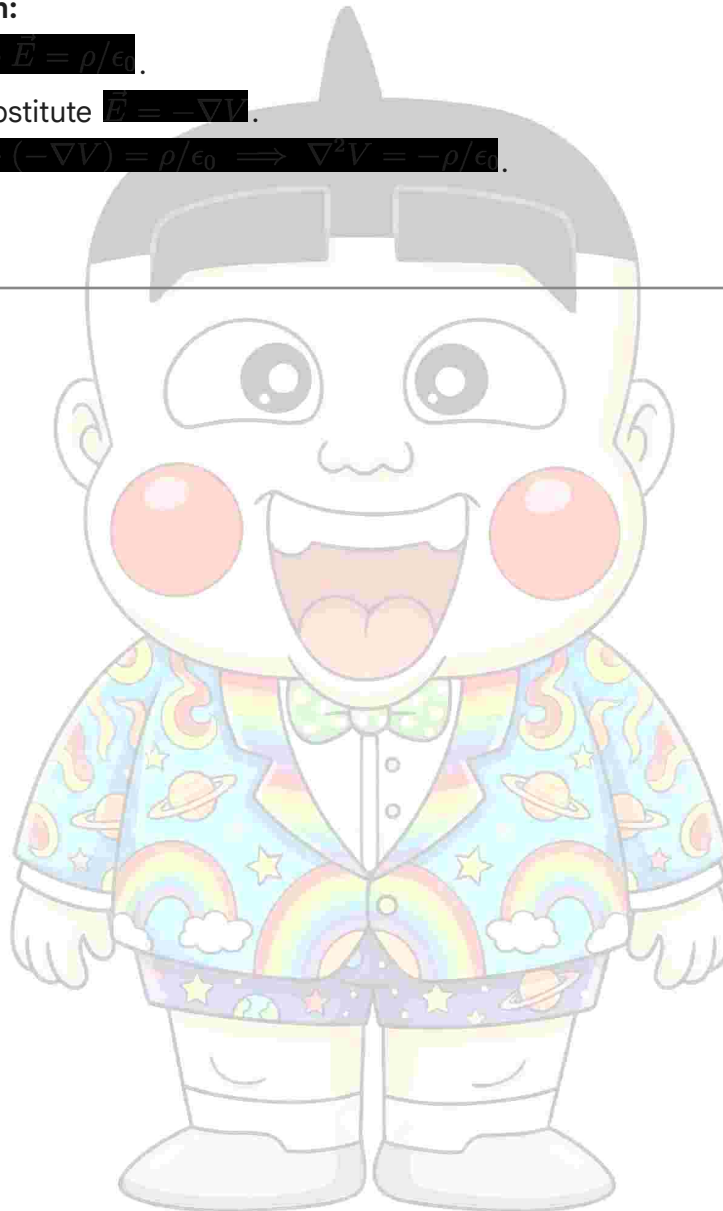
48. Biot-Savart Law & Application

- **Question:** State Biot-Savart law (vector form). Find $\vec{B}$ at the center of a circular loop.

- Answer: $\vec{B} = \frac{\mu_0 I}{2R}$.

49. Poisson's Equation

- **Question:** Derive Poisson's equation from Gauss law.
- **Derivation:**
 - $\nabla \cdot \vec{E} = \rho/\epsilon_0$.
 - Substitute $\vec{E} = -\nabla V$.
 - $\nabla \cdot (-\nabla V) = \rho/\epsilon_0 \implies \nabla^2 V = -\rho/\epsilon_0$.



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